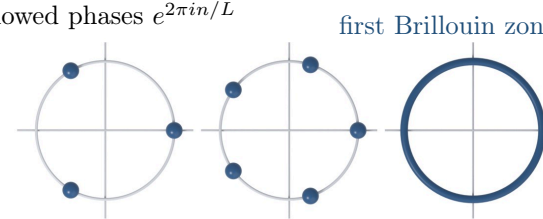

$\alpha_j \propto e^{2\pi i n j/3}$
$n = 0$
$n = 1$
$n = 2$


allowed phases $e^{2\pi i n/L}$
first Brillouin zone
$L = 3$
$L = 5$
$L \to \infty$